

Classifying LLM Prompt Intent Using Sentence Embeddings

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Abstract

This project trains a four-class text classifier to identify the intent of prompts submitted to large language models (LLMs): creative, informational, task-oriented, or adversarial. The classifier uses frozen sentence embeddings from ‘all-MiniLM-L6-v2’ as input features, with three classification heads compared — Logistic Regression, Linear SVM, and MLP. The dataset is assembled from three public Hugging Face sources, hand-curated seeds and synthetic examples generated via the Anthropic API (Claude). The best model reaches approximately 82% accuracy and 0.82 macro F1, and is deployed as a Gradio demo on Hugging Face Spaces.

1 Introduction

The rapid deployment of LLMs in consumer and enterprise products has made prompt classification an active area of both research and engineering. Knowing the intent behind a user prompt before it reaches the model enables smarter routing, pre-filtering of adversarial requests, and analytics over usage patterns. Current production systems typically rely on keyword lists or regular expressions — techniques that are brittle in the face of semantically creative circumvention.

This project applies sentence embeddings to the intent classification problem. The hypothesis is that intent is a semantic property of a prompt, and that a dense vector representation will capture distributional differences between intent classes that a sparse, lexical approach cannot. This is particularly important for the adversarial class: a jailbreak that wraps a harmful request in creative fiction shares semantic structure with other adversarial prompts even when it shares no keywords with them (Perez and Ribeiro, 2022).

2 Dataset

The dataset combines three public sources with hand-curated seeds. **WildChat-1M** (Zhao et al., 2024) contains real user–ChatGPT conversations; first-turn user messages are extracted and labeled heuristically using keyword patterns (e.g. “write” → creative, “explain” → informational, “fix” → task) as proxies for intent, since no ground-truth labels are available. **lmsys/toxic-chat** (Lin et al., 2023) provides real user prompts rated for toxicity by human annotators — prompts marked toxic are mapped to the adversarial class. **walledai/AdvBench** (Zou et al., 2023) is a benchmark of adversarial attack prompts, all labeled adversarial.

Twenty hand-written seed examples per class anchor the distribution, and synthetic augmentation via the Anthropic API (Claude) generates additional balanced examples. The final dataset contains 867 examples split 80/20 into 693 training and 174 test examples with stratified sampling. The resulting class distribution is reasonably balanced, ranging from 183 to 240 examples per class (task: 183, adversarial: 219, creative: 225, informational: 240). A source field is preserved for per-origin analysis.

Label ambiguity is an acknowledged limitation: a creative prompt that incidentally requests technical detail may sit on the boundary between creative and task. The model predicts the dominant intent rather than all applicable intents.

3 Method

3.1 Embeddings

All prompts are encoded with sentence-transformers/all-MiniLM-L6-v2 (Wang et al., 2020), a 22M-parameter BERT variant distilled for semantic sentence similarity. It produces 384-dimensional unit-normalised vectors and runs efficiently on CPU. Encoder weights

are frozen; only the classification head is trained, keeping training time under one minute.

3.2 Classifiers

Three classifiers are compared, each preceded by StandardScaler normalisation:

Classifier	Key configuration
Logistic Regression	C=5, balanced weights
Linear SVM	C=1, balanced weights
MLP	(512, 256), ReLU, early stop

Table 1: Classifier configurations.

class_weight=“balanced” is applied because adversarial prompts are naturally underrepresented in real prompt distributions. Without balancing, classifiers optimise accuracy by ignoring the minority class.

3.3 Evaluation

Models are evaluated on accuracy, macro F1, and per-class F1. A confusion matrix identifies systematic error patterns — specifically whether adversarial prompts are misclassified as creative.

4 Results

Classifier	Acc.	F1 mac.	F1 wt.
Logistic Regression	0.822	0.822	0.821
Linear SVM	0.782	0.782	0.782
MLP (512, 256)	0.810	0.809	0.810

Table 2: Classifier comparison on held-out test set.

Logistic Regression achieves the best scores, suggesting the embedding space is already largely linearly separable — the frozen all-MiniLM-L6-v2 representations cluster intent classes clearly enough that a linear decision boundary suffices. Table 3 shows the per-class breakdown for the best model.

Class	Precision	Recall	F1
creative	0.78	0.89	0.83
informational	0.84	0.77	0.80
task	0.80	0.89	0.85
adversarial	0.87	0.75	0.80

Table 3: Per-class results for Logistic Regression (best model).

The adversarial class shows the lowest per-class recall (0.75), consistent with the hypothesis that

creative-framed jailbreaks are the hardest to classify correctly — the model achieves high precision (0.87) but misses 25% of true adversarial prompts. The task and creative classes achieve the highest recall (0.89), reflecting that these intent types have the most distinct linguistic patterns in the embedding space.

5 Deployment

The best model is serialised with joblib and uploaded to belrem/llm-prompt-intent-classifier on Hugging Face Hub. A Gradio 5 interface at <https://huggingface.co/spaces/belrem/llm-prompt-intent-demo> accepts any prompt and returns the predicted intent class, confidence bars across all four classes, and a contextual explanation.

Reflection on AI Tools

Claude was used throughout this project for scaffolding the dataset pipeline, writing the training loop, building the Gradio interface, and drafting this report. The code generated was largely correct but required active verification at every step. The dataset loading logic for WildChat initially used a deprecated API argument; the Gradio interface had an output mismatch that only surfaced during local testing; and the synthetic generation script referenced a non-existent model version. Each required cross-referencing documentation and independent debugging.

More subtly, Claude’s first draft of the model card overstated the adversarial classifier’s reliability, requiring human judgment to reframe it appropriately. The main productivity gain was eliminating boilerplate, freeing time for the more important decisions: class definitions, handling class imbalance, and interpreting results. AI tools are most valuable when they handle the predictable and the human handles the uncertain.

Limitations

Adversarial classification is inherently uncertain — this classifier is a heuristic signal, not a security guarantee. The dataset (693 examples) is small by NLP standards, and the WildChat heuristic labeling introduces noise. Intent can be ambiguous, and jailbreak patterns evolve faster than static training sets.

Artifacts

- **Code:** <https://github.com/BelenRemedi/llm-prompt-intent-classifier>
- **Dataset:** <https://huggingface.co/datasets/belrem/llm-prompt-intent>
- **Demo:** <https://huggingface.co/spaces/belrem/llm-prompt-intent-demo>

References

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